

Project Report

ME2010 Engineering Innovation and Design

Submitted by: Group A09
Registration No: E/22/395, E/22/399, E/22/401, E/22/410,
E/22/423

Faculty of Engineering
University of Peradeniya

Submitted: 15th April 2026

Contents

1	Introduction	2
2	Opportunity Identification	2
2.1	Methodology	2
3	Customer Needs Analysis	3
3.1	List of Customer Needs	3
3.2	AHP Priority Vector	3
4	Product Specification	4
5	Concept Generation	4
6	Concept Selection	5
6.1	Technical Design & Assembly	5
6.2	2D Assembly Drawing	6
6.3	2D Part Drawings	6
7	Conclusions	7
8	References	7
A	Appendix	7

1 Introduction

The Universal Mechanical Brace is an assistive orthopedic device designed to provide stabilization and pain relief for individuals suffering from joint discomfort, specifically targeting the knees and lower back.

In Sri Lanka, musculoskeletal disorders are highly prevalent among manual laborers and elderly populations. This project aims to deliver a cost-effective and locally manufacturable alternative to expensive imported solutions. Currently available commercial products are often priced out of reach for the average rural worker, and cheaper alternatives lack the necessary structural support and longevity.

Key Objectives:

- Provide effective structural support
- Enable universal adaptability (5th–95th percentile users)
- Maintain affordability for local accessibility

2 Opportunity Identification

The opportunity was identified through observation of common physical strain in agricultural and industrial workers.

2.1 Methodology

- Target Group: Ages 18–60
- Survey Method: Digital questionnaire

Key Insight: 72% of participants reported experiencing body pain and fatigue even after light work.

3 Customer Needs Analysis

3.1 List of Customer Needs

The following needs were identified from the customer need analysis to ensure the brace meets user expectations.

Need No.	Customer Need	Category
1	The brace is lightweight and comfortable for long-duration wear.	Comfort
2	The brace provides effective mechanical stabilization for the knee and spine.	Support
3	The brace is affordable for elderly users and those in rehabilitation.	Mobility
4	The brace is easily adjustable to accommodate different body sizes.	Ease of use
5	The brace assists in maintaining correct posture during heavy lifting.	Adjustability
6	The brace is portable and easy to store when not in use.	Portability

Table 1: List of Customer Needs

3.2 AHP Priority Vector

Customer Need	Priority (%)
Support and Effectiveness	33.73
Comfort	23.42
Ease of Use / Mobility	14.68
Adjustability	8.59

Table 2: Normalized Priority Vector

4 Product Specification

Specifications were derived by mapping customer needs to engineering metrics.

The target specifications ensure the mechanical brace meets the identified customer needs while remaining within technical and economic constraints.

Metric No.	Need Nos.	Metric	Importance	Unit	Approx. Value
1	1, 3	Total mass	4	Kg	< 2.0
2	2, 5	Support Load Capacity	5	N	> 750
3	4	Range of Motion (Knee)	5	Degrees	0–140
4	2, 5	Spinal Alignment Deviation	5	Degrees	< 3
5	1	Surface Pressure (on skin)	4	kN/m ²	< 4.5
6	3	Unit manufacturing cost	5	Rs	< 5000
7	1	Breathability (Airflow rate)	3	m ³ /s	> 0.05
8	4	Number of adjustment points	3	Count	4–6
9	1	Time to put on/take off	3	s	< 60
10	6	Folded volume (Portability)	2	cm ³	< 5000

Table 3: Product Target Specifications

5 Concept Generation

To address the customer needs, the conceptualization phase began with a functional decomposition. A function diagram was developed to map the core system inputs—such as the misaligned user limb, input energy, and joint angle signals—to the desired outputs, which include a re-aligned limb, corrective helping torque, and stability feedback.

Based on this functional understanding, five distinct concepts were generated, each exploring different methods of structural support and actuation:

- Spring Brake Hybrid (SBO):** This concept utilizes a mechanical hinge featuring an assist spring designed to provide energy return, coupled with a friction brake/damper for user control and joint locking. The primary struts are AL-6061-T6 aluminum, with cuffs made of Nylon 12 and Carbon Fiber.
- Pneumatic Artificial Muscle (PAM):** An active support system utilizing an air pump and PVC air tubes to drive pneumatic artificial muscles. The structural casing is constructed from an aluminum alloy and ABS plastic.
- Active Cable Suit:** A tension-based bracing system featuring an elastic core integrated with isoprene rubber tubing. The cables run through stainless steel pulleys and are precisely tensioned using a 10cm rotating buckle.

4. **Adaptive Wedge Orthosis (AWO):** This concept focuses on dynamic flexibility using an internal variable-density elastomer wedge mechanism. Housed in a molded carbon fiber composite frame, it uses thermoformed EVA memory foam condyle pads to adapt to the user's load and positioning
5. **Spring Hinged Mechanical Brace (SHM):** A purely mechanical approach centered on an aluminum hinge integrated with a spring, providing resistive or assistive force. It utilizes adjustable velcro straps and is lined with cotton and foam cushioning to ensure user comfort.

6 Concept Selection

Based on the Quality Function Deployment (QFD) analysis and concept ranking matrix, the **Adaptive Wedge Orthosis** was ranked as the primary solution, with the Spring Brake Hybrid serving as a strong mechanical alternative. The final design integrates a mechanical hinge system that restricts extension beyond 180° to prevent hyperextension injuries.

6.1 Technical Design & Assembly

The following figures illustrate the finalized 3D CAD model of the Adaptive Wedge Orthosis. Figure 3 displays the complete final assembly alongside an exploded view detailing the internal mechanism and structural components.



Figure 1: Final Assembly View

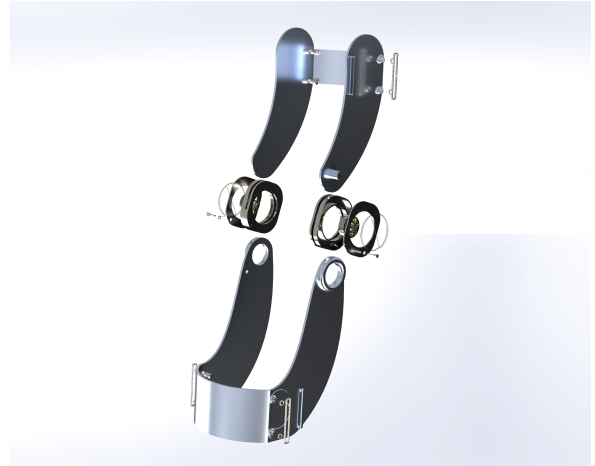


Figure 2: Exploded View of Components

Figure 3: 3D CAD Models of the Finalized Mechanical Brace

6.2 2D Assembly Drawing

To facilitate manufacturing and establish exact geometric tolerances, a comprehensive 2D assembly drawing was generated. Figure 04 details the standard orthographic projections.

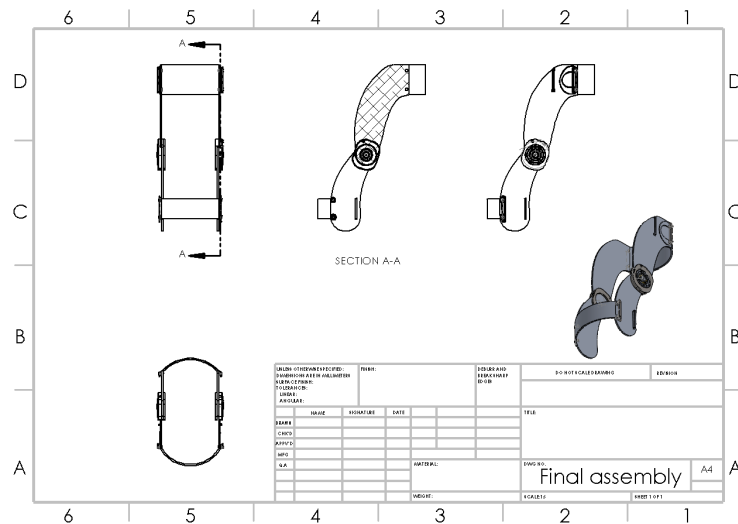


Figure 4: Assembly drawing and projections

6.3 2D Part Drawings

Due to the extensive number of individual custom components requiring detailed geometric dimensioning and tolerancing, the comprehensive set of 2D manufacturing drawings has been compiled into a dedicated external repository.

Manufacturing Data Repository:

[Click Here to View 2D Part Drawings \(Google Drive\)](#)

These part drawings include exact dimensional specifications, material callouts, and surface finish tolerances required for the local fabrication of the brace components.

7 Conclusions

This project successfully addresses a critical gap in affordable orthopedic support devices in Sri Lanka.

- Scientifically driven design process
- Balanced performance and comfort
- Scalable and manufacturable locally

8 References

1. University of Peradeniya – ME2010 Materials
2. [Survey Data](#)
3. Ulrich & Eppinger – Product Design and Development

A Appendix

The following figures provide the full analytical matrices and charts used during the conceptualization and evaluation phases of the mechanical brace design.

			Items											
			1	2	3	4	5	6						
			Support and Effectiveness	Comfort	Ease of use	Mobility	Adjustability	Portability	Row Total	Relative importance	Scaled Importance			
Items	1	Support and Effectiveness	1	2	3	3	4	5	18.00	33.73%	33.73	2.227183804	6.123049	
	2	Comfort	0.5	1	2	2	3	4	12.50	23.42%	23.4236	1.409052653	6.116514	
	3	Ease of use	0.333	0.5	1	1	2	3	7.83	14.68%	14.6782	0.827230442	6.06482	
	4	Mobility	0.333	0.5	1	1	2	3	7.83	14.68%	14.6782	0.827230442	6.06482	
	5	Adjustability	0.25	0.333	0.5	0.5	1	2	4.58	8.59%	8.58803	0.489073116	6.02552	
	6	Portability	0.2	0.25	0.333	0.333	0.5	1	2.62	4.90%	4.90209	0.31369476	6.040675	
	Column Sum		2.616	4.583	7.833	7.833	12.5	18				6.072567		
			Normalized Matrix						Row AVG	Percentage		CI	0.014513	
			0.38226	0.4364	0.383	0.383	0.32	0.27778	0.36374	36.37%		CR	0.011704	
			0.19113	0.2182	0.25533	0.25533	0.24	0.22222	0.23037	23.04%				
			0.12729	0.1091	0.12767	0.12767	0.16	0.16667	0.1364	13.64%				
			0.12729	0.1091	0.12767	0.12767	0.16	0.16667	0.1364	13.64%				
			0.09557	0.07266	0.06383	0.06383	0.08	0.11111	0.08117	8.12%				
			0.07645	0.05455	0.04251	0.04251	0.04	0.05556	0.05193	5.19%				

Figure 5: Analytic Hierarchy Process (AHP) Matrices for Customer Needs

Rank the Concepts				PRODUCT	MECHANICAL BRACE
				DATE	24/02/2026
Selection Criteria	SBO (Ref)	PAM Support	Active Cable Suit	TCO Hinge	Adaptive Wedge
Support Effectiveness	0	+	+	+	+
Comfort	0	-	+	0	+
Ease of Use	0	-	0	0	+
Mobility	0	+	+	0	+
Adjustability	0	0	-	+	+
Portability	0	0	-	+	+
Visual Appeal	0	-	+	0	+
Environmental effects	0	-	0	+	+
Sum +s	0	2	4	4	8
Sum 0's	8	2	2	4	0
Sum -s	0	4	2	0	0
Net Score	0	-2	2	4	8
Rank	4	5	3	2	1

Figure 6: Concept Ranking and Selection Matrix

Quality Function Deployment									
<									

Figure 7: Quality Function Deployment (QFD)